

# Analyzing VLTI/GRAVITY Data of $\beta$ Pictoris b for Exomoon-Induced Exoplanetary Wobble

Saji Alqadhi, Emma Zheng, Madeleine Morse-Hill, Thao Nhi Nguyen, Arbar Sharif

## Abstract

Detecting exomoons would improve our understanding of planetary system formation and refine exosolar models, but it is a major observational challenge in modern astronomy. With significantly fainter apparent magnitudes and smaller gravitational effects compared to their host planets, no exomoons have been confirmed to exist. In this study, we analyze VLTI/GRAVITY astrometric time series data for super-Jupiter  $\beta$  Pic b for induced reflex motion to search for a potential exomoon. We applied polynomial detrending and subtracted of  $\beta$  Pic c's effects to increase data precision. After simulating and subtracting a regular Keplerian orbit for  $\beta$  Pic b, we analyzed the residuals for wobble motion but were unable to conclusively determine the presence of an exomoon due to lack of data precision.

## Introduction

Exomoons are moons orbiting planets beyond our solar system. In our own system, moons are responsible for shaping planetary evolution by stabilizing axial tilts, influencing the lengths of the day, and stabilizing a planet's ring and satellites. Our Moon stabilizes Earth's axial tilt, which gives us a stable climate and seasons, and its gravity creates the ocean tides. In addition, moons can generate tidal heating; Jupiter's Europa and Saturn's Enceladus have subsurface oceans warmed by tidal friction generated due to the gravitational pull from their host planet (Benz-Narveson, 2025). This heating mechanism can provide the potential to sustain life on a moon or a planet lacking heat energy from a sun; this includes potential exomoons outside our solar system (Tokadjian & Piro, 2023, p. 1). This suggests that detecting and characterizing these moons would not only help refine exo-solar system models to help us better understand their formations, but also increase our count of potentially habitable worlds beyond traditional Earth-like planets. Observational data will also reveal whether large, stable moons are a common byproduct of planet formation or a rare anomaly of our Solar System.

As of late 2025, over 6,000 exoplanets have been confirmed, yet no exomoon has been conclusively detected (NASA Exoplanet Archive, 2025). As Kipping (2009) explains, this is due to a combination of inherent physical limitation and observational challenges that produce false-positive results; exomoons are substantially lesser in size, brightness, and gravitational effects than their host planets, making them extremely difficult to observe or detect at vast interstellar distances. Past attempts have provided some evidence of the potential existence of exomoons; currently, the exomoon candidates with the greatest supporting evidence include

Kepler-1625 b-i and Kepler-1708 b-i. Teachey & Kipping (2018) proposed a Neptune-size moon orbiting Kepler-1625 b after finding support from a transit timing variation and a photometric dip. However, this evidence is not widely accepted; Heller & Hippke (2024) claimed the evidence for Kepler-1625 b-i was shallow, since it only looked at one transit, and that upon reanalyzing data from Hubble, Kepler-1625 b-i had much less statistical evidence supporting its existence than Teachey & Kipping (2018) had proposed.

The most promising proposed exomoon detection methods include radial-velocity observation, direct transit observation, and astrometric time series observation. Ruffio et al. (2023) demonstrates the effectiveness of radial-velocity observations to detect exomoons around self-luminous planets, but  $\beta$  Pic b is not self-luminous. Direct transit observation has been used in the past, in studies such as Teachey & Kipping (2018), but is very difficult to use since moons vary in relative position to their planet from transit to transit (Szabo et al., 2010, p. 1). In addition, these former exomoon detection techniques are hindered by stellar noise, limited resolution, and alignment constraints that produce ambiguous or unreliable results (Kammerer et al., 2025).

VLTI offers a superior approach by utilizing K-band interferometry to perform astrometric time series observations, which penetrates dust and reduces stellar glare (NASA, 2006), producing high angular resolutions as precise as  $50 \mu\text{as}$ . Much of obstructive stellar dust is penetrated since K-band wavelengths are large enough to pass through the dust, and stellar glare is reduced since most stars (including  $\beta$  Pictoris A) peak in radiation in the UV to visible light range, making it easier to detect smaller objects around it (such as planets) when utilizing a near-infrared wavelength. VLTI/GRAVITY works by combining light beams from four Unit Telescopes (each with a primary mirror diameter of 8.2 meters), as well as some Auxiliary Telescopes (1.2 primary mirror diameter) to generate interference fringe patterns that allow for extremely precise data (ESO, 2023). The telescopes work together to act as one giant “virtual” telescope of  $\sim 200$  meters in diameter, which according to Winterhalder et al. (2025) can detect exoplanet wobbles due to a moon at least around the size of a gas giant ( $\sim 0.14 M_{\text{Jup}}$ ). VLTI observes an exoplanet’s motion and exomoons influence directly instead of guessing with transit observations, making it an effective yet relatively unexplored method for searching for exomoons.

The  $\beta$  Pictoris system is very young, at around 20–26 million years old, and it is located  $\sim 63$  light years away from our solar system.  $\beta$  Pic b is a super-Jupiter ( $11.729 M_{\text{Jup}}$ ) that was first announced in 2009 based on imaging from 2003, making it one of the first exoplanets ever to be directly imaged. It orbits its A-type host star,  $\beta$  Pictoris ( $1.7 M_{\odot}$ ), at about 10 AU (outside habitable zone) and has an orbital period of about 23 years, allowing ample time for observation. Its orbit has an eccentricity of  $\sim 0.1$ , and its spin is extremely fast ( $\sim 8$  hours per rotation). We chose  $\beta$  Pic b because it has been identified as one of the most promising exoplanets for evaluating for a potential exomoon in the short-term (Kral et al., 2025) due to its relatively

abundant data availability from VLTI/GRAVITY and likelihood of having an exomoon. This likelihood is due to its severe axial tilt ( $\sim 60$  degrees) despite its high spin angular momentum, which suggests a strong and stabilizing force on the planet which could be best explained by an exomoon (Poon et al., 2024). However, influences from its star as well as  $\beta$  Pic c, the other exoplanet in the system (located much closer to the star), could complicate our wobble detection process.

Other exoplanets such as Kepler-1625 b and Kepler-1708 b may be more well-known as promising candidates for having exomoons, but they are both located in the northern celestial hemisphere which VLTI/GRAVITY cannot observe.

## Technical Details

Exoplanetary reflex motion, also known as a wobble, is detected as an exoplanet and its potential exomoon orbit their shared barycenter, which produces a measurable change in the planet’s position relative to its host star (Winterhalder et al., 2025). Studies show that GRAVITY, the VLTI’s upgraded interferometric instrument, can reach a miniscule precision of detecting  $50 \mu\text{as}$  orbital displacements, making it one of the few facilities capable of directly observing exoplanetary motion instead of inferring from photometric data (Lacour et al., 2025; NASA, 2006). A  $50 \mu\text{as}$  wobble could be caused by a moon around the size of a gas giant orbiting an even larger host planet.  $\beta$  Pic b is estimated to have the mass of a super-Jupiter ( $\sim 9.3 M_{\text{Jup}}$  from our posterior), making it a strong candidate for hosting a detectable moon-induced wobble.

VLTI/GRAVITY data for  $\beta$  Pic b is publicly available through the ESO Archive (ESO, 2020). We selected 9 epochs spanning approximately 6 months, from September 2019 to March 2020, chosen for their relatively even temporal spacing and data quality. These epochs are listed in Table 1. A full list of all available VLTI/GRAVITY observations of  $\beta$  Pic b, including epochs outside our selected window, is provided in Appendix A.

**Table 1:** Selected GRAVITY dates of observation of  $\beta$  Pic b.

| Date of Observation |
|---------------------|
| 2019-09-12          |
| 2019-11-09          |
| 2019-11-11          |
| 2020-01-07          |
| 2020-02-09          |
| 2020-02-10          |
| 2020-02-12          |
| 2020-03-08          |
| 2020-03-12          |

## Data Reduction and Cleaning

Raw VLTI/GRAVITY observations are stored as interferometric fringe data in FITS format and cannot be used directly for astrometry. We used two separate software tools to convert this raw data into usable RA/Dec positional offsets: **EsoRex** for initial pipeline reduction, and **ExoGravity** for astrometric extraction and detrending. We first installed and ran the ESO GRAVITY Pipeline’s **EsoRex** along with the associated calibration data to convert each raw interferometric epoch into reduced FITS files. **EsoRex** handles all instrument-level corrections including baseline geometry calibration, detector dark and flat corrections, and fringe pattern extraction, producing intermediate pipeline products for each epoch. All 9 epochs were successfully reduced through **EsoRex**. However, **EsoRex** does not output RA/Dec positional offsets directly; it only produces the reduced fringe data that the next step requires.

We then used **ExoGravity**, from the GRAVITY Consortium Python Tools, to extract the scientifically meaningful astrometric solution from each reduced `dualscivis.fits` file. **ExoGravity** fits the planet’s position within the fringe pattern using  $\chi^2$  minimisation, producing RA and Dec offsets in milliarcseconds relative to the host star. This step also performs polynomial detrending to remove slowly varying stellar background contamination and reduce atmospheric and seismic noise, using polynomial degrees of 3, 4, and 6 as in the method used in Kral et al. (2025). For most epochs we used the standard `poly=4` setting, which is the default in **ExoGravity** and the value used in the original ExoGRAVITY papers for  $\beta$  Pic b. The sole exception was the 2020-03-08 epoch, for which `poly=4` produced near-zero, degenerate formal errors; we therefore used `poly=6` for that epoch, which gave a more stable and physically sensible astrometric solution.

One epoch, 2019-11-09, was excluded from our analysis entirely after reduction. The **ExoGravity** reduction for this epoch produced unusable output as the extracted astrometric solution was unreliable and inconsistent with all other epochs. This left us with 8 usable epochs for subsequent orbit modelling. For epochs with multiple exposures taken at different baseline configurations (specifically 2020-02-12 and 2020-03-08, each of which produced two exposures labelled p0 and p1), we used the longest-baseline exposure per epoch as a conservative single-file reduction. This is consistent with the approach of Kral et al. (2025) and other ExoGRAVITY papers for epochs with low planet contrast, where the longest baseline provides the most precise fringe measurement. Since the  $\beta$  Pic system is nearly edge-on, the RA and Dec positional offsets are highly correlated, a small change in the assumed orbital geometry produces strongly covarying shifts in both coordinates simultaneously.

In our **ExoGravity** output, the correlation coefficient  $\rho$  between the RA and Dec offset uncertainties reached  $\rho = +1$  for epochs 2020-02-10 and 2020-03-08 p0, and  $\rho = -1$  for epoch 2020-03-12. A correlation coefficient of

exactly  $\pm 1$  is numerically degenerate and would cause something called a matrix inversion error in Orbitize!’s likelihood calculation. We therefore manually adjusted these three values to  $\rho = \pm 0.99$  which would preserve the near-perfect correlation while avoiding the numerical singularity. This is a standard pragmatic fix for this class of near-edge-on astrometric data and does not materially affect the orbital fit given the already extreme correlation. The final cleaned astrometric dataset, consisting of RA/Dec offsets and associated uncertainties for each usable epoch, is presented in Table 2 below. An important thing to note is how most epochs produced a nan in the error column. This means ExoGravity did not produce formal error estimates when completing the polynomial detrending.

**Table 2:** RA/Dec offsets and errors for each epoch after ExoGravity reduction, excluding the unusable 2019-11-09 epoch.

| Epoch # | Date          | MJD      | RA (mas) | Dec (mas) | Error                   |
|---------|---------------|----------|----------|-----------|-------------------------|
| 1       | 2019-09-12    | 58738.27 | 24.11    | 5.36      | nan                     |
| 3       | 2019-11-11    | 58798.38 | −2.23    | 20.45     | nan                     |
| 4       | 2020-01-07    | 58855.21 | 6.63     | 4.20      | nan                     |
| 5       | 2020-02-09    | 58888.17 | 35.81    | 10.09     | nan                     |
| 7       | 2020-02-12 p0 | 58891.08 | 43.49    | 16.64     | nan                     |
| 8       | 2020-02-12 p1 | 58891.09 | −0.04    | 0.41      | nan                     |
| 9       | 2020-03-08 p0 | 58916.07 | 8.05     | 47.18     | $\pm 1.07$ / $\pm 2.35$ |
| 10      | 2020-03-08 p1 | 58916.07 | 9.13     | 44.56     | $\pm 1.01$ / $\pm 0.15$ |
| 11      | 2020-03-12    | 58920.08 | −1.21    | 17.37     | nan                     |

## Orbit Modelling

We used *Orbitize!*, an open source python orbit-fitting package (Blunt et al., 2020), to perform Keplerian orbit modelling for the entirety of the  $\beta$  Pic system encompassed in our data. It applies Bayesian inference methods to fit orbital parameters based on astrometric data. Using this approach allows us to estimate not only a single best-fit orbit, but a full distribution of possible orbits that are all reasonably consistent with our observations.

In *Orbitize!*, we implemented a Markov Chain Monte Carlo (MCMC) which is an algorithm that uses parallel tempered ensemble samplers, which are advanced algorithms designed to sample from complex, high-dimensional probability distributions (Foreman-Mackey et al., 2013; Vousden et al., 2015). This helps us avoid getting stuck at a value and provides us with a wide range of potential orbits to analyze. We used the default parameters in our initial testing, but quickly found that we needed to tailor our MCMC configuration to account for the reality of our data set discussed below. This was done to ensure the MCMC did a sufficient amount of exploration of our parameter space to end up with more stable posterior distributions, while tuning

the number of temperatures to improve the mixing between chains. Our initial one-planet fits to  $\beta$  Pic b alone attempted to explain astrometric measurements that contained positional information from both  $\beta$  Pic b and  $\beta$  Pic c using a single Keplerian orbit. Because  $\beta$  Pic c’s gravitational influence on the star’s barycentric motion is non-negligible, it tried to modulate the apparent sky position of  $\beta$  Pic b. Only fitting only one planet produced unphysical posteriors and poorly constrained orbit plots. Our solution was to construct a combined relative-astrometry input file using `num_secondary_bodies = 2` and `fit_secondary_mass = True` to indicate our data was looking at both objects relative to the star. This is important because *Orbitize!*’s multi-planet mode accounts for how planet c perturbs the star’s barycentric motion, which in turn affects the apparent motion of planet b. While not a full N-body integration by default, this is a much more sensible baseline model than a single-planet fit. For epochs with multiple exposures, we adopted the longest-baseline exposure per epoch as a conservative single-file reduction, consistent with the approach of Kral et al. (2025) and other ExoGRAVITY papers for epochs with low planet contrast.

For the 2020-03-08 epoch specifically, we used `poly=6` in the exogravity reduction rather than the standard `poly=4` we used for the rest. We did this because `poly=4` produced near-zero, degenerate formal errors for that epoch; `poly=6` gave a more stable and physically sensible astrometric solution. All other epochs used `poly=4` because they produced sensible error free results and `poly=4` was the default from exogravity and we only felt the need to deviate when our data required it.

We also had to manually adjust priors on all orbital and system parameters to ensure our MCMC explored a physically realistic parameter space that is consistent and reasonable considering the published properties of the objects in the system:  $\beta$  Pic b of course but also  $\beta$  Pic c. All priors we decided on were chosen to reflect realistic orbital configurations without overly restricting the model simulation. We had to choose between Uniform and Gaussian priors depending on what knowledge we could find from published research. We used Uniform priors for parameters when we wanted to remain uninformative or agnostic across a specific range of values. We used Gaussian priors when we had somewhat precise external physical measurements for certain parameters so *Orbitize!* could create probabilities assuming our value input was the most likely while in uniform priors we essentially didn’t have a certain value we were confident was the most accurate and let *Orbitize!* generate one from our data. For example, according to a combination of radial velocity and direct imaging astronomy, in Lagrange et al. (2020) (A&A 642, A18), the semi-major axis of  $\beta$  Pic b is reported as  $9.8 \pm 0.4$  AU (and up to 10.3 AU when including absolute astrometry from Hipparcos and Gaia which is why we decided to use that value). In Nowak et al. (2020) (A&A 642, A2), the semi-major axis of the companion planet,  $\beta$  Pic c, is reported as  $2.7 \pm 0.02$  AU. All parameter values were pulled from sources we managed to scrape from official published research but when errors were high and there was conflict, we

opted to choose a reasonable uniform prior instead of a gaussian prior which assumes a higher certainty. The rest of our priors were pulled from Nielsen et al. (2014) and Brandt et al. (2021).

The MCMC output took into account our priors and then considered them with our data and produced distributions for each of the orbital parameters. It gave us the median value for the best estimate with all intervals being credible, we used the default 68% range of confidence to represent our uncertainty.  $\beta$  Pic b is expected to have a near-circular orbit due to disk interactions. Nowak et al. (2020) report  $e < 0.1$  at  $3\sigma$ , and Brandt et al. (2021) report  $e = 0.119 \pm 0.008$  from combined astrometry. An upper bound of 0.35 prevents the sampler from exploring high-eccentricity solutions while keeping some cushion room open. For inclination, the  $\beta$  Pic debris disk is nearly perfectly edge-on. Millar-Blanchaer et al. (2015) (ApJ 811, 18) measure a disk inclination of  $88.9 \pm 0.2^\circ$ , and the planet’s inclination is consistent with the disk plane. A  $1.5^\circ$  sigma is conservative. For the position angle of nodes ( $\Omega$ ), we used `GaussianPrior(212.0°,  $\sigma = 2.5^\circ$ )` because it follows the disk position angle. Millar-Blanchaer et al. (2015) found a disk PA  $\sim 29\text{--}31^\circ$ . In *Orbitize!*, this convention corresponds to  $\sim 211\text{--}212^\circ$ . Without this prior the sampler would explore all  $360^\circ$  freely so we wanted to constrain it. Which is what we ended up doing for the other priors. The argument of periastron ( $\omega$ ) was left at the default `UniformPrior(0,  $2\pi$ )` because it’s not independently constrained from any literature for our short arc. Similarly, the epoch of periastron passage ( $\tau$ ) with `UniformPrior(0, 1)` is also not constrained by our short arc, which is why we left it at default.

We also did the same process of parameter search for  $\beta$  Pic c. For the semimajor axis we used `GaussianPrior(2.7 AU,  $\sigma = 0.3$  AU)` because  $\beta$  Pic c’s semimajor axis is well-measured at  $\sim 2.68\text{--}2.75$  AU (Nowak et al., 2020). A 0.3 AU sigma allows reasonable exploration around this value while trusting its accuracy. For the eccentricity we ended up using a `UniformPrior(0.0, 0.5)`. Recent literature values for  $\beta$  Pic c’s eccentricity range from  $\sim 0.2$  to  $0.3$  (Brandt et al., 2021). A uniform prior from 0.0 to 0.5 spans this range without artificially truncating the posterior by assuming a single likely value and moving around it. Using our data, the eccentricity median was calculated at 0.283 which is incredibly trustworthy given the range of possible eccentricities. For inclination, we used `GaussianPrior(89.0°,  $\sigma = 2.0^\circ$ )`.  $\beta$  Pic c is inferred to orbit in the same near-edge-on plane as  $\beta$  Pic b and the disk so their inclination which is relative to our perspective is pretty much the same. We used a  $2.0^\circ$  sigma which is slightly more generous than for b because of our greater uncertainty for  $\beta$  Pic c. For the position angle of nodes we used `GaussianPrior(211.0°,  $\sigma = 3.0^\circ$ )` (Millar-Blanchaer et al., 2015). It’s consistent with the disk PA and the inclination of  $\beta$  Pic b. A  $3.0^\circ$  sigma is wider than for b, again because of our uncertainty for  $\beta$  Pic c due to its closer position to the star and how uncertain literature we’ve looked though described it. Finally for the argument of periastron and the epoch of periastron passage, we just kept the default uniform prior because there was no published literature we

could find on these values and we didn't want to potentially skew the results by changing it.

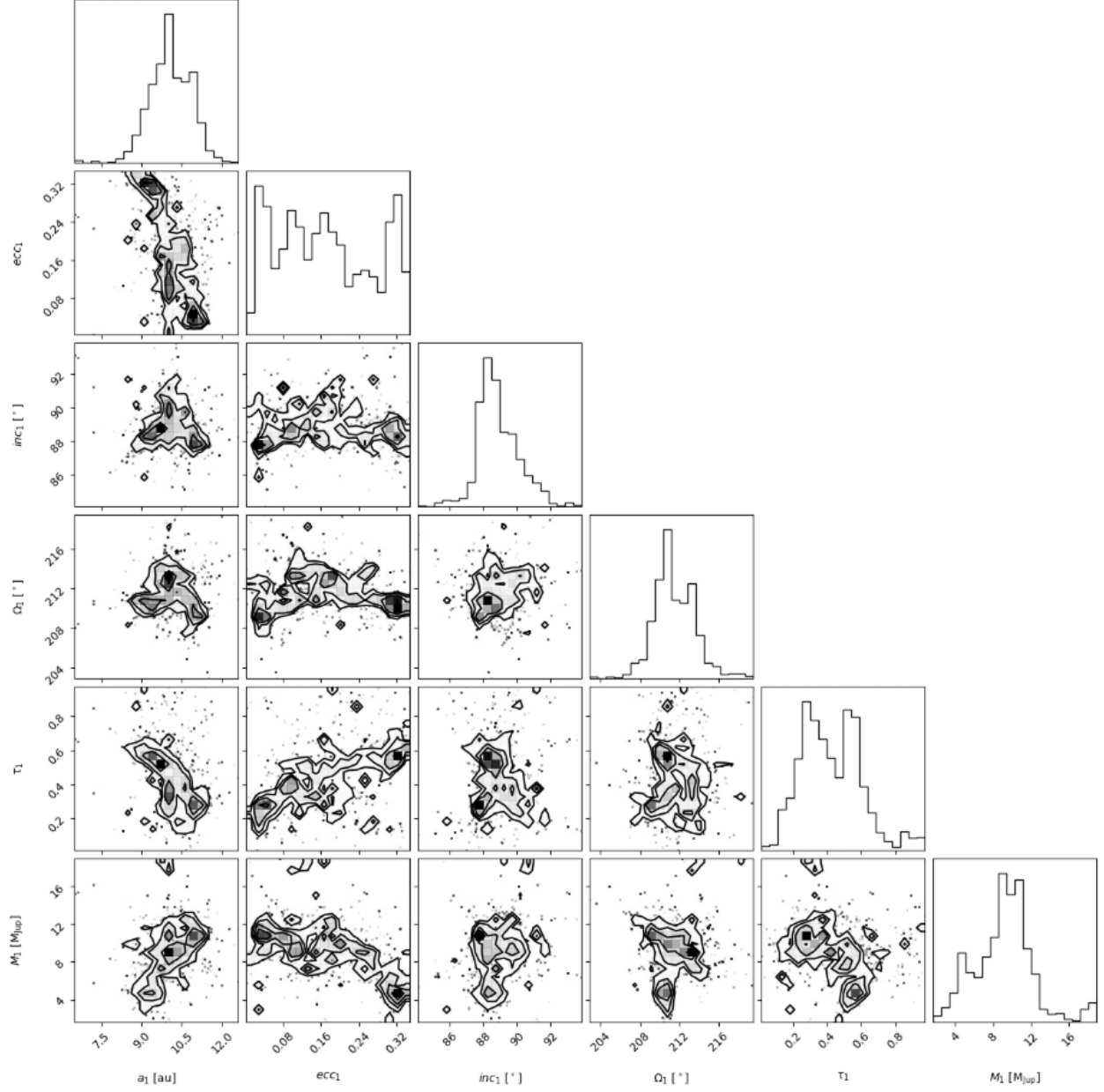
The last portion of our priors was made for the shared system parameters. For parallax, we chose 51.44 mas with  $\sigma = 0.12$  mas from van Leeuwen (2007), the Hipparcos reduction, confirmed by Gaia DR3. Accounting for the entire system's parallax could be incredibly useful for making sure our data is correctly aligned. The next three measurements were concerning masses of all objects in the system to potentially account for the gravitational effects they exert on each other. Stellar mass ( $m_0$ ) had **GaussianPrior**( $1.75 M_{\odot}$ ,  $\sigma = 0.05 M_{\odot}$ ) from Crifo et al. (1997) and confirmed by multiple spectroscopic analyses; Nowak et al. (2020) adopt  $1.75 M_{\odot}$  and is fairly recent especially alongside our data. For the two object masses, we used Brandt et al. (2021). For  $\beta$  Pic b mass ( $m_1$ ), we set **GaussianPrior**( $9.8 M_{\text{Jup}}$ ,  $\sigma = 3.0 M_{\text{Jup}}$ ) with our broad prior centered on  $\sim 9.8 M_{\text{Jup}}$  with a  $3.0 M_{\text{Jup}}$  sigma; not so tight as to dominate the astrometric fit but wide enough to give us some space around our value to calculate probabilities for. Finally,  $\beta$  Pic c mass ( $m_2$ ) has a **GaussianPrior**( $8.9 M_{\text{Jup}}$ ,  $\sigma = 1.5 M_{\text{Jup}}$ ) We had a broad prior centered on  $\sim 8.9 M_{\text{Jup}}$  with a  $1.5 M_{\text{Jup}}$  sigma. Lastly, the orbital period ( $P$ ) and mutual inclination, no priors were set because we wanted to derive them from our data along with the other priors. We're pleased with the results because the generated values align with published values (Vandal et al., 2020).

A table of our priors and posteriors can be seen below:

**Table 3:** Priors and posteriors for the two-planet *Orbitize!* fit. Posteriors are quoted as median with 68% credible intervals.

| Quantity                   | Prior                                  | Posterior (median, 68% CI)     |
|----------------------------|----------------------------------------|--------------------------------|
| $a_b$ (AU)                 | Gaussian(10.3, 1.0)                    | $10.074^{+0.839}_{-0.723}$     |
| $e_b$                      | Uniform(0, 0.35)                       | $0.158^{+0.143}_{-0.109}$      |
| $i_b$ (deg)                | Gaussian( $88.9^\circ$ , $1.5^\circ$ ) | $88.638^{+1.423}_{-0.817}$     |
| $\omega_b$ (deg)           | Uniform(0, $2\pi$ )                    | $159.315^{+77.538}_{-71.809}$  |
| $\Omega_b$ (deg)           | Gaussian( $212^\circ$ , $3^\circ$ )    | $211.205^{+2.228}_{-1.458}$    |
| $\tau_b$                   | Uniform(0, 1)                          | $0.402^{+0.179}_{-0.148}$      |
| $a_c$ (AU)                 | Gaussian(2.7, 0.3)                     | $2.568^{+0.221}_{-0.320}$      |
| $e_c$                      | Uniform(0, 0.5)                        | $0.283^{+0.090}_{-0.084}$      |
| $i_c$ (deg)                | Gaussian( $89^\circ$ , $2^\circ$ )     | $87.217^{+3.777}_{-3.110}$     |
| $\omega_c$ (deg)           | Uniform(0, $2\pi$ )                    | $232.847^{+25.006}_{-133.874}$ |
| $\Omega_c$ (deg)           | Gaussian( $211^\circ$ , $3^\circ$ )    | $208.130^{+3.143}_{-3.318}$    |
| $\tau_c$                   | Uniform(0, 1)                          | $0.461^{+0.307}_{-0.087}$      |
| Parallax (mas)             | Gaussian(51.44, 0.12)                  | $51.389^{+0.160}_{-0.116}$     |
| $M_b$ ( $M_{\text{Jup}}$ ) | Gaussian(10, 3)                        | $9.302^{+2.203}_{-3.738}$      |
| $M_c$ ( $M_{\text{Jup}}$ ) | Gaussian(9, 2)                         | $8.784^{+0.626}_{-1.301}$      |
| $M_\star$ ( $M_\odot$ )    | Gaussian(1.75, 0.05)                   | $1.728^{+0.064}_{-0.024}$      |
| $P_b$ (yr)                 | derived                                | $24.130^{+3.268}_{-2.373}$     |
| $P_c$ (yr)                 | derived                                | $3.110^{+0.392}_{-0.560}$      |
| Mutual inc. (deg)          | derived                                | $4.817^{+2.405}_{-2.529}$      |

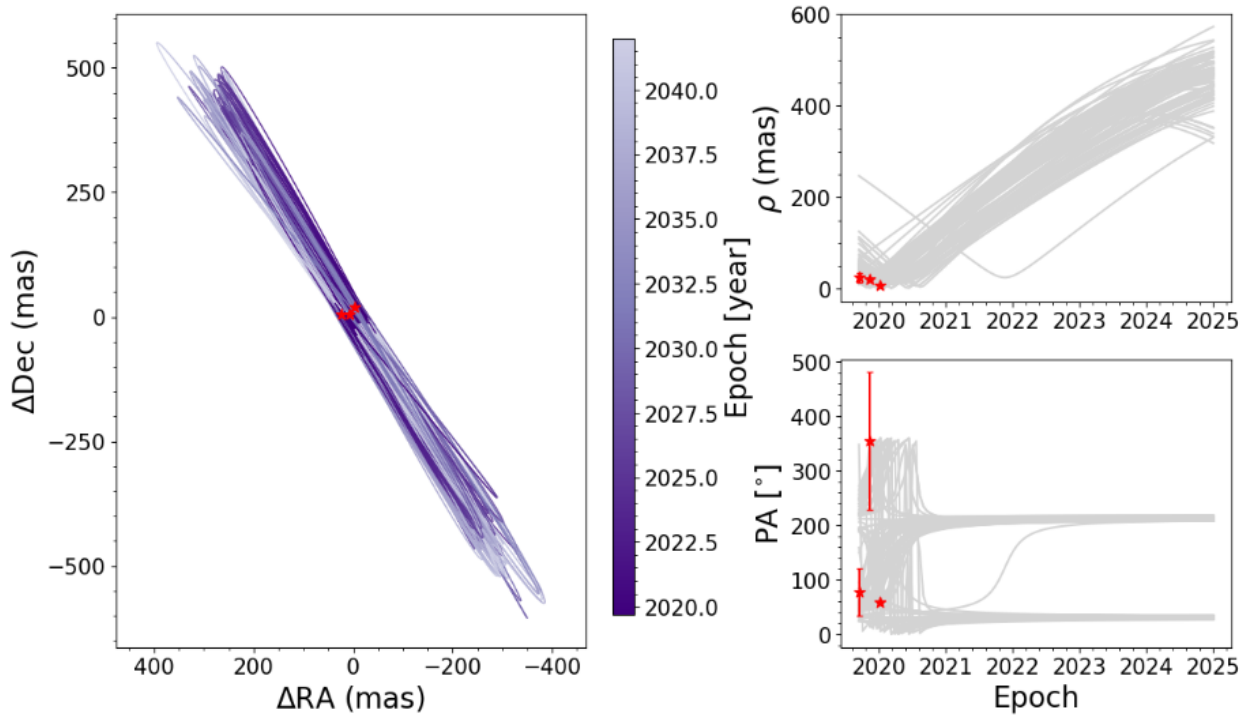
To help us visualize our results, we generated corner and orbit plots for  $\beta$  Pic b and  $\beta$  Pic c by separating the data for each object. An important thing to note is that we ran the MCMC with 5 temps, 50 walkers with 10,000 total orbits and we used that as our base run since a larger run done after produced smear corner plots and the orbit plot was chaotic. *Orbitize!* shows us our sample orbital solutions overlaid on our observed astrometric data (Figures 1, 2, 3, and 4).



**Figure 1:** Corner plot of the posterior distributions for  $\beta$  Pic b orbital parameters from our two-planet Orbitize! MCMC run (5 temperatures, 50 walkers, 10,000 total orbits). Diagonal panels show marginalised 1D posteriors; off diagonal panels show joint 2D distributions. Parameters shown are semimajor axis  $a_1$ , eccentricity  $e_1$ , inclination  $i_1$ , longitude of ascending node  $\Omega_1$ , epoch of periastron  $\tau_1$ , and planet mass  $M_1$ .

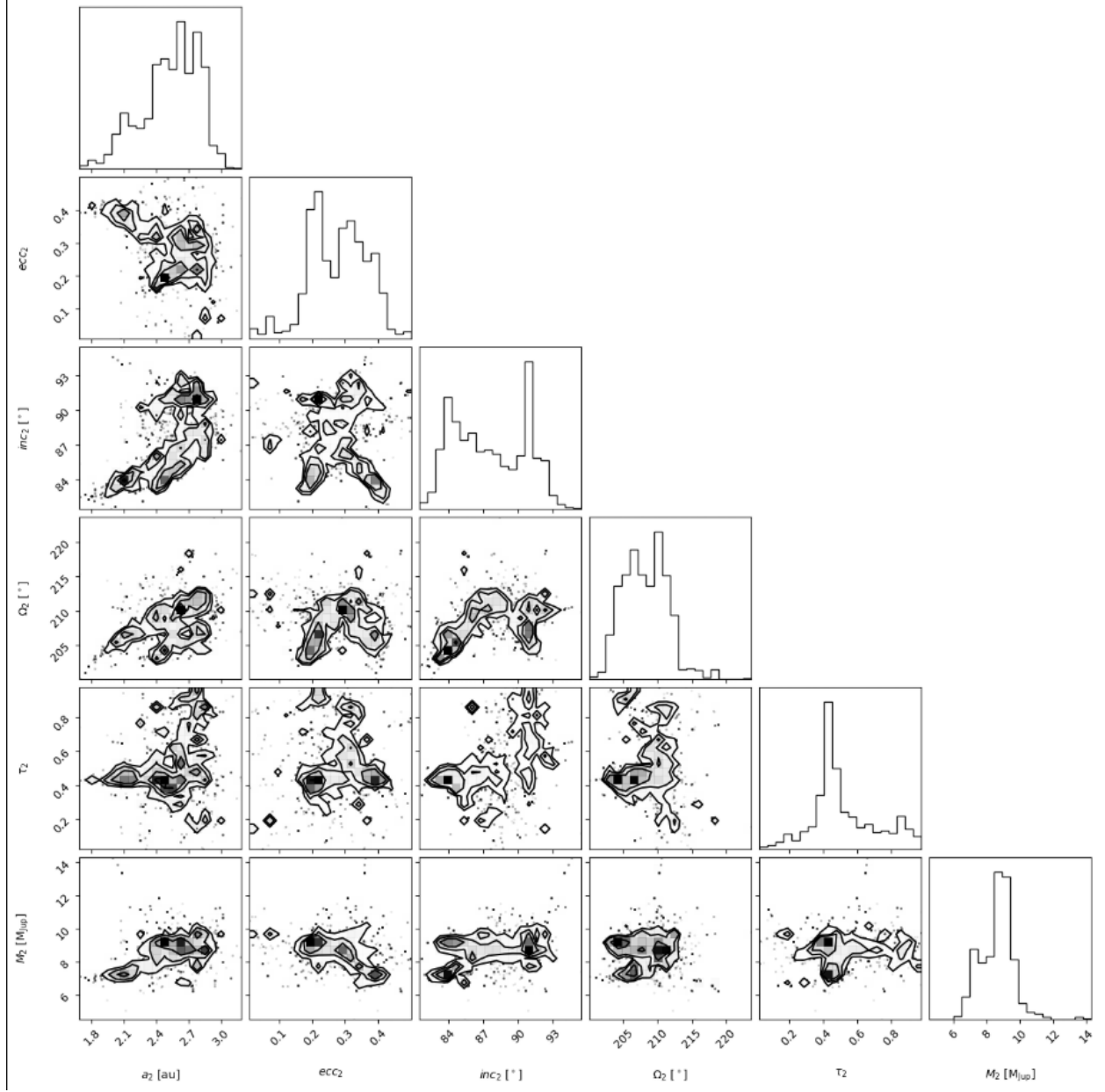
The corner plot for  $\beta$  Pic b (Figure 1) shows the marginalised one-dimensional posterior distributions along the diagonal and the joint two-dimensional distributions in the off-diagonal panels for the six fitted parameters: semimajor axis  $a_1$ , eccentricity  $e_1$ , inclination  $i_1$ , longitude of ascending node  $\Omega_1$ , epoch of periastron  $\tau_1$ , and planet mass  $M_1$ . The semimajor axis posterior is well-peaked between approximately 9.5 and 10.5 AU, consistent with our prior centered at 10.3 AU and with literature values from Lagrange et al. (2020) and

Nowak et al. (2020). The inclination and  $\Omega_1$  posteriors are both sharply peaked with  $\text{inc}_1$  near  $88\text{--}89^\circ$  and  $\Omega_1$  near  $211\text{--}212^\circ$  and both are visibly prior-dominated, reflecting that our short arc contains little independent leverage on these angles beyond what the disk geometry already tells us (Millar-Blanchaer et al., 2015). The eccentricity posterior is broad and relatively flat across the full allowed range of  $\sim 0.08\text{--}0.32$ , indicating that our six epochs are insufficient to independently constrain the orbital shape; the posterior median of 0.158 is consistent with Brandt et al. (2021) but the wide 68% credible interval of  $(-0.109, +0.143)$  reflects this degeneracy. The epoch of periastron  $\tau_1$  is the least constrained parameter, showing a broad, multi-modal distribution with concentrations near 0.2–0.3 and 0.5–0.6, which is expected given the short arc and the near-circular orbit. The planet mass  $M_1$  is broadly distributed from  $\sim 4$  to  $\sim 14 M_{\text{Jup}}$ , peaking near  $8\text{--}10 M_{\text{Jup}}$ , consistent with our wide prior of  $9.8 \pm 3.0 M_{\text{Jup}}$ . In the 2D panels, the most prominent correlation is between  $a_1$  and  $\text{ecc}_1$ , where larger semimajor axes are preferred alongside higher eccentricities. A well-known degeneracy in orbit fitting from short astrometric arcs, where a more eccentric orbit on a wider path can produce similar projected separations to a more circular orbit on a smaller one. The spread in the orbit ensemble and the broad posteriors on  $\text{ecc}_1$ ,  $\tau_1$ , and  $M_1$  collectively confirm that while our baseline orbit is physically sensible and consistent with the literature, additional epochs spanning a longer arc would be required to meaningfully tighten these parameters.



**Figure 2:** Orbit plot for  $\beta$  Pic b showing the posterior ensemble of orbital solutions in the sky plane ( $\Delta\text{RA}$  vs  $\Delta\text{Dec}$ ), colour-coded by epoch from 2020 to 2040 (darker purple indicates earlier epochs with higher sampling density). Red stars indicate observed VLTI/GRAVITY data points. Right panels show angular separation  $\rho$  (top) and position angle PA (bottom) as functions of time.

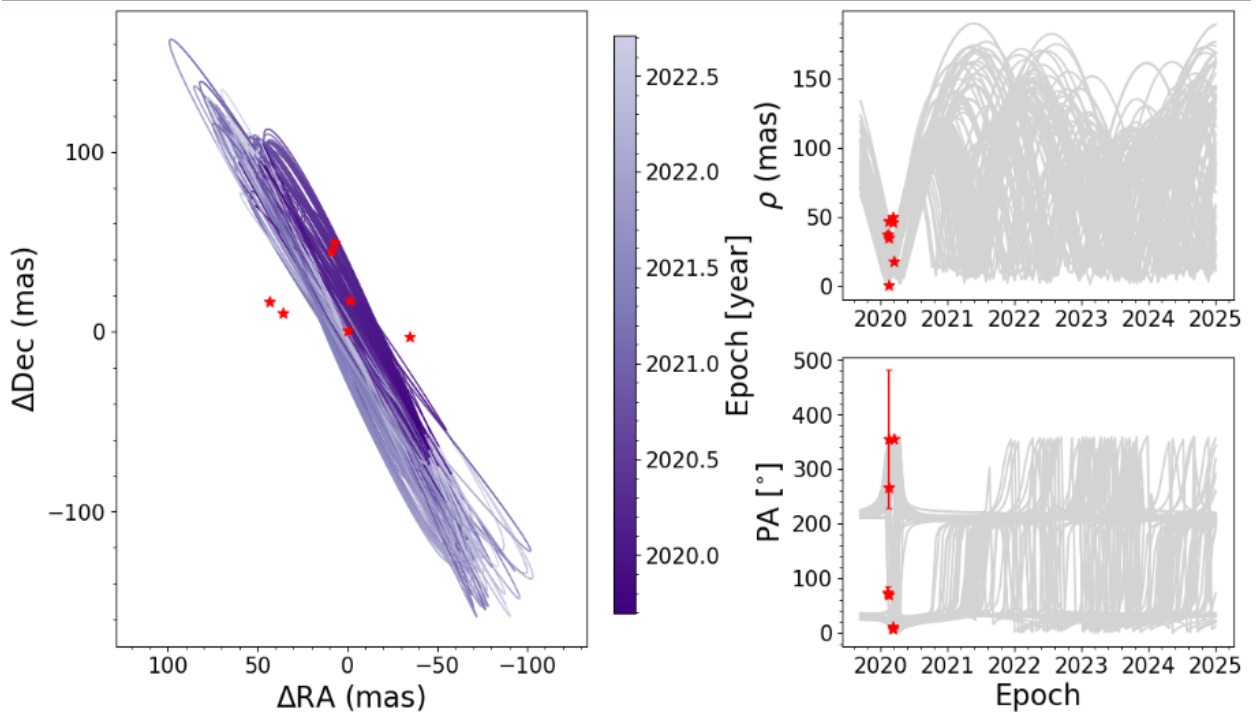
The orbit plot for  $\beta$  Pic b (Figure 2) displays the full ensemble of posterior orbit samples in the sky plane ( $\Delta\text{RA}$  vs  $\Delta\text{Dec}$ ), colour-coded by epoch from 2020 to 2040 with the darker purple having more certainty. Our six observed data points, shown as red stars, are tightly clustered near the origin which is consistent with the fact that our observations span only a  $\sim 6$ -month arc of a  $\sim 24$ -year orbit, so the planet moves very little in angular separation relative to the star within our window. The right panels show separation ( $\rho$ ) and position angle (PA) as functions of time. The separation panel shows the orbit ensemble fanning out from near-zero in 2020 to a wide spread of roughly 200–550 mas by 2025, reflecting genuine uncertainty in the semimajor axis and eccentricity given our short observational arc. The PA panel shows the observed points scattered across a wide range ( $\sim 50^\circ$ – $350^\circ$ ) in 2020, consistent with the poorly constrained argument of periastron  $\tau_1$  and  $\omega_1$  at this phase of the orbit, before the ensemble converges toward a stable  $\sim 210$ – $220^\circ$  by 2022 as the Gaussian prior on  $\Omega_1$  pulls the chains toward the known disk position angle.



**Figure 3:** Corner plot of the posterior distributions for  $\beta$  Pic c orbital parameters from our two-planet Orbitize! MCMC run. Parameters shown are semimajor axis  $a_2$ , eccentricity  $e_2$ , inclination  $i_2$ , longitude of ascending node  $\Omega_2$ , epoch of periastron  $\tau_2$ , and planet mass  $M_2$ .

The corner plot for  $\beta$  Pic c (Figure 3) shows several notable differences from  $\beta$  Pic b. The semimajor axis posterior  $a_2$  is bimodal, with two distinct peaks near  $\sim 2.2$  AU and  $\sim 2.7$  AU, suggesting that two families of orbital solutions remain consistent with our data which likely corresponds to orbits sampled at different phases of the planet’s periastron passage within our window. The eccentricity  $ecc_2$  is more clearly peaked than  $ecc_1$  was, concentrated between  $\sim 0.2$  and  $0.3$  and consistent with our posterior median of  $0.283$ , reflecting that the shorter orbital period allows the data to place slightly more constraint on the orbital shape. The

inclination posterior  $\text{inc}_2$  is bimodal, with two distinct peaks near  $\sim 86^\circ$  and  $\sim 90^\circ$ , wider than the sharp single peak seen for  $\beta$  Pic b; this is consistent with the larger prior uncertainty ( $\sigma = 2.0^\circ$  vs  $1.5^\circ$  for b) and the greater sensitivity of a shorter-period edge-on orbit to small inclination changes. The  $\Omega_2$  posterior shows two concentrations near  $\sim 208^\circ$  and  $\sim 213^\circ$ , slightly less tightly constrained than  $\Omega_1$ , again reflecting the greater difficulty of pinning down the nodal geometry for the inner, faster-moving planet. In strong contrast to  $\beta$  Pic b, the epoch of periastron  $\tau_2$  is sharply peaked near 0.4–0.5, making it the best-constrained angular parameter for  $\beta$  Pic c; this is because our 6-month arc covers a meaningful fraction of the 3.1-year orbit, allowing the data to identify where within the orbital cycle the planet was observed. The planet mass  $M_2$  is more tightly distributed than  $M_1$ , with a sharp peak near 7–8  $M_{\text{Jup}}$  and a posterior median of 8.784  $M_{\text{Jup}}$ , consistent with our prior of  $8.9 \pm 1.5 M_{\text{Jup}}$ . In the 2D panels, the  $a_2$  to  $\text{ecc}_2$  correlation persists as in  $\beta$  Pic b, but with more structured two-cloud morphology reflecting the bimodal semimajor axis solution. The  $\text{inc}_2$  to  $\text{ecc}_2$  panel also shows a more pronounced correlation than seen for b, with the two inclination families preferring slightly different eccentricities, further evidence of the orbital degeneracies introduced by the near-edge-on geometry.



**Figure 4:** Orbit plot for  $\beta$  Pic c showing the posterior ensemble of orbital solutions in the sky plane ( $\Delta\text{RA}$  vs  $\Delta\text{Dec}$ ), colour-coded by epoch from 2020 to 2022.5. Red stars indicate observed VLTI/GRAVITY data points. The compact overlapping loops reflect multiple full orbital cycles within the projection window, consistent with  $\beta$  Pic c’s short orbital period of  $\sim 3.1$  yr.

The orbit plot for  $\beta$  Pic c (Figure 4) reveals a markedly different character from  $\beta$  Pic b, reflecting the much

shorter orbital period of  $\sim 3.1$  years. The sky-plane ensemble shows compact, overlapping elliptical loops rather than the long diagonal arcs of  $\beta$  Pic b, because multiple full orbital cycles occur within the 2020–2025 projection window. Crucially, the eight observed data points (red stars) are spread more visibly around the orbital arc rather than clustered near a single point, because our 6-month observational window covers a substantially larger fraction of  $\beta$  Pic c’s orbit ( $\sim 16\%$ ) than it does of  $\beta$  Pic b’s ( $\sim 2\%$ ). The separation panel shows the observed points at  $\sim 10$ – $50$  mas, with the orbit ensemble showing repeated sharp dips toward zero separation at each periastron passage and recovering to  $\sim 150$  mas at apoastron, consistent with the more eccentric posterior ( $e_c = 0.283$ ). The PA panel is markedly more chaotic than for  $\beta$  Pic b, with chains oscillating wildly across  $0^\circ$ – $500^\circ$  — this is a direct consequence of the near-edge-on inclination ( $\sim 87^\circ$ ) causing the position angle to undergo a near- $180^\circ$  flip near each conjunction, making PA extremely sensitive to the exact orbital phase and eccentricity sampled. The complexity of the  $\beta$  Pic c orbit plot is a direct consequence of its shorter period ( $\sim 3.1$  yr) and near-edge-on geometry, which cause PA to oscillate rapidly and make the orbital solution sensitive to small changes in eccentricity and inclination.

## Results & Discussion

We did not conclusively detect an exomoon around Beta Pic b by looking for an exoplanetary wobble using currently available interferometry data. Below we report the residual astrometry from our two-planet baseline fit and compare it quantitatively to the expected scale of an exomoon-induced wobble signal.

After subtracting the best-fit two-planet Keplerian baseline orbit from our observed VLTI/GRAVITY astrometry, we computed the residual displacement  $|d|$  at each epoch for both  $\beta$  Pic b and  $\beta$  Pic c. These residuals were taken from our base MCMC run with 5 temps, 50 walkers, with a total of 10,000 orbits as it produced better-converged posteriors than the larger run. The results of the residuals are as follows:

$$\begin{aligned} \beta \text{ Pic b: } & \text{median } |d| = 30.049 \text{ mas, } \quad \max |d| = 42.796 \text{ mas} \\ \beta \text{ Pic c: } & \text{median } |d| = 26.306 \text{ mas, } \quad \max |d| = 57.998 \text{ mas} \end{aligned}$$

The residuals for  $\beta$  Pic b (presented visually in Figures 1 & 2), show no coherent periodic pattern consistent with a reflex wobble; the data points scatter across a wide range in both RA and Dec with no clear elliptical structure around the origin that would be expected from a moon-induced astrometric signal. To assess whether our residuals could plausibly contain an exomoon signal, we compare them to the expected astrometric reflex amplitude derived from the Kral et al. (2025) analytic scaling. If a moon of mass  $M_{\text{moon}}$  orbits a planet of

mass  $M_p$  at semimajor axis  $a_{\text{moon}}$ , the planet’s reflex semimajor axis is:

$$a_{\text{ast}} \approx a_{\text{moon}} \frac{M_{\text{moon}}}{M_p + M_{\text{moon}}} \approx q a_{\text{moon}} \quad (1)$$

and the peak-to-peak angular displacement on the sky is:

$$\Delta \approx \frac{2 a_{\text{ast}}}{d} \quad (2)$$

where  $d$  is the distance to the system. For  $\beta$  Pic b, adopting  $d \approx 19.4$  pc (from our fitted parallax of 51.389 mas),  $M_p \approx 9.3 M_{\text{Jup}}$ , and a hypothetical moon at  $a_{\text{moon}} \sim 0.04$  AU which is the dynamically relevant range of 0.03–0.05 AU predicted by Poon et al. (2024) the expected signals are either a 1 Neptune-mass moon: reflex semimajor axis  $\sim 12 \mu\text{as}$ , peak-to-peak wobble  $\sim 24 \mu\text{as}$  or a 10 Neptune-mass moon: reflex semimajor axis  $\sim 120 \mu\text{as}$ , peak-to-peak wobble  $\sim 240 \mu\text{as}$ .

The physically interesting exomoon signal for  $\beta$  Pic b therefore lies in the range of tens to a few hundreds of microarcseconds which is the  $\mu\text{as}$  to sub-mas regime in which GRAVITY operates (Kral et al., 2025; Winterhalder et al., 2025). Our measured  $\beta$  Pic b residuals, with a median of 30.049 mas, are roughly two orders of magnitude larger than this expected signal. To put it simply: 30 mas equals 30,000  $\mu\text{as}$  which is more than 1,000 times the  $\sim 24 \mu\text{as}$  wobble expected from a Neptune-mass moon, and more than 100 times the  $\sim 240 \mu\text{as}$  wobble expected from a 10 Neptune-mass moon. This comparison is the quantitative basis for our non-detection. The subtracted best-fit orbital model leaves residuals far too large to search for a moon signal within them. In the HD 206893 exomoon search of Kral et al. (2025), the candidate astrometric signal corresponded to a reflex radius of only  $\sim 0.14$ – $0.15$  mas in the RA-Dec residual plane, and even that was treated as only tentative because instrumental systematics can also produce small elliptical patterns around the origin. Our residuals are more than two orders of magnitude larger than that candidate signal. These large residuals are best interpreted as reflecting baseline-model mismatch rather than instrumental noise alone. With only three  $\beta$  Pic b epochs spanning six months of a  $\sim 24$ -year orbit, the orbital parameters (particularly the eccentricity and epoch of periastron) remain poorly constrained in our posterior, meaning the baseline orbit itself carries substantial uncertainty. This propagates directly into the residuals.

It is important to note that this does not constitute evidence against the existence of an exomoon around  $\beta$  Pic b; it indicates that the precision of our current baseline model is insufficient to isolate the  $\sim \mu\text{as}$ -scale signal that an exomoon would produce. A longer observational baseline with additional epochs would be required to reduce residuals to a level where a meaningful moon search becomes possible.

## Acknowledgements

We would like to thank the other astronomy lab group that uses the Deimos computer for letting us hog it for a few weeks during our initial data preparation process and Dr. Linch for his support and advice throughout this process.

## Bibliography

### References

- Benz-Narveson, S. (2025). *Tidal heating in gas giant ice moons*. Senior Seminars and Capstones, 11. <https://digitalcommons.morris.umn.edu/capstone/11>
- Blunt, S., Wang, J. J., Angelo, I., et al. (2020). *orbitize! : A comprehensive orbit-fitting software package for the high-contrast imaging community*. *The Astronomical Journal*, 159(3), 89. <https://doi.org/10.3847/1538-3881/ab6663>
- Brandt, G. M., Brandt, T. D., Dupuy, T. J., Li, Y., & Michalik, D. (2021). *Precise dynamical masses and orbital fits for  $\beta$  Pic b and  $\beta$  Pic c*. *The Astronomical Journal*, 161(4), 179. <https://doi.org/10.3847/1538-3881/abdc2e>
- Brandt, T. D. (2021). *The Hipparcos–Gaia catalog of accelerations: Gaia EDR3 edition*. *The Astrophysical Journal Supplement Series*, 254(2), 42. <https://doi.org/10.3847/1538-4365/abf93c>
- Crifo, F., Vidal-Madjar, A., Lallement, R., Ferlet, R., & Gerbaldi, M. (1997).  *$\beta$  Pictoris revisited by Hipparcos: Star properties*. *Astronomy & Astrophysics*, 320, L29–L32. <https://ui.adsabs.harvard.edu/abs/1997A%26A...320L..29C/abstract>
- ESO Archive. (2020). *Observational raw data query interface*. Retrieved October 15, 2025, from [https://archive.eso.org/eso/eso\\_archive\\_main.html](https://archive.eso.org/eso/eso_archive_main.html)
- ESO. (2023, July 26). *The very large telescope interferometer*. Retrieved October 15, 2025, from <https://www.eso.org/sci/facilities/paranal/telescopes/vlti.html>
- Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. (2013). *emcee: The MCMC hammer*. *Publications of the Astronomical Society of the Pacific*, 125(925), 306–312. <https://doi.org/10.1086/670067>
- Heller, R., & Hippke, M. (2024). *Large exomoons unlikely around Kepler-1625 b and Kepler-1708 b*. *Nature Astronomy*, 8, 193–206. <https://doi.org/10.1038/s41550-023-02148-w>

- Kammerer, J., Eisenhauer, F., Nowak, M., Lacour, S., Pfuhl, O., Gillessen, S., & GRAVITY+ Collaboration. (2025). *Exoplanet detection and astrometry with the VLTI/GRAVITY+ instrument* [Preprint]. arXiv. <https://arxiv.org/pdf/2509.15304>
- Kenworthy, M. A., Mellon, S. N., Bailey, J. I., et al. (2021). *The  $\beta$  Pictoris b Hill sphere transit campaign*. *Astronomy & Astrophysics*, 648, A15. <https://doi.org/10.1051/0004-6361/202040060>
- Kipping, D. M. (2009). *Transit timing effects due to an exomoon*. *Monthly Notices of the Royal Astronomical Society*, 392(1), 181–189. <https://doi.org/10.1111/j.1365-2966.2008.13999.x>
- Kral, Q., Wang, J., Kammerer, J., Lacour, S., Malin, M., Winterhalder, T., et al. (2025). *Exomoon search with VLTI/GRAVITY around the substellar companion HD 206893 B*. *Astronomy & Astrophysics*. <https://doi.org/10.48550/ARXIV.2511.20091>
- Lacour, S., Carrión-González, Ó., & Nowak, M. (2025). *Exoplanets in reflected starlight with dual-field interferometry*. *Astronomy & Astrophysics*, 694, A277. <https://doi.org/10.1051/0004-6361/202451060>
- Lagrange, A.-M., Rubini, P., Nowak, M., et al. (2020). *Unveiling the  $\beta$  Pictoris system, revisiting the mass and flagging a potential third planet with GRAVITY*. *Astronomy & Astrophysics*, 642, A18. <https://doi.org/10.1051/0004-6361/202038823>
- Millar-Blanchaer, M. A., Graham, J. R., Pueyo, L., et al. (2015).  *$\beta$  Pictoris’ inner disk in polarized light and new orbital parameters for  $\beta$  Pictoris b*. *The Astrophysical Journal*, 811(1), 18. <https://doi.org/10.1088/0004-637X/811/1/18>
- National Aeronautics and Space Administration. (2006). *Infrared interferometry technology development for exoplanet detection* (NASA Technical Report No. 20060014034). NASA Technical Reports Server. <https://ntrs.nasa.gov/api/citations/20060014034/downloads/20060014034.pdf>
- NASA Exoplanet Archive. (2025, October 30). Retrieved October 31, 2025, from <https://exoplanetarchive.ipac.caltech.edu/>
- Nielsen, E. L., Liu, M. C., Wahhaj, Z., et al. (2014). *The Gemini NICI planet-finding campaign: The frequency of giant planets around young B and A stars*. *The Astrophysical Journal*, 794(2), 158. <https://doi.org/10.1088/0004-637X/794/2/158>
- Nowak, M., Lacour, S., Lagrange, A.-M., et al. (2020). *Peering into the formation history of  $\beta$  Pictoris b with VLTI/GRAVITY long baseline interferometry*. *Astronomy & Astrophysics*, 642, A2. <https://doi.org/10.1051/0004-6361/202038325>

- Poon, M., Rein, H., & Pham, D. (2024). *A potential exomoon from the predicted planet obliquity of  $\beta$  Pictoris b*. *The Open Journal of Astrophysics*, 7. <https://doi.org/10.33232/001c.127130>
- Ruffio, J.-B., Horstman, K., Mawet, D., et al. (2023). *Detecting exomoons from radial velocity measurements of self-luminous planets: Application to observations of HR 7672 B and future prospects*. ArXiv. <https://doi.org/10.48550/ARXIV.2301.04206>
- Szabo, G. M., Simon, A. E., Kiss, L. L., & Regaly, Z. (2010). *Practical suggestions on detecting exomoons in exoplanet transit light curves*. ArXiv. <https://doi.org/10.48550/ARXIV.1012.1560>
- Teachey, A., & Kipping, D. M. (2018). *Evidence for a large exomoon orbiting Kepler-1625b*. *Science Advances*, 4(10), eaav1784. <https://doi.org/10.1126/sciadv.aav1784>
- Tokadjian, A., & Piro, A. L. (2023). *Tidal heating of exomoons in resonance and implications for detection*. *The Astronomical Journal*, 165(4), 173. <https://doi.org/10.3847/1538-3881/acc254>
- van Leeuwen, F. (2007). *Validation of the new Hipparcos reduction*. *Astronomy & Astrophysics*, 474(2), 653–664. <https://doi.org/10.1051/0004-6361:20078357>
- Vandal, T., Rameau, J., & Doyon, R. (2020). *Dynamical mass estimates of the  $\beta$  Pictoris planetary system through Gaussian process stellar activity modelling*. *The Astronomical Journal*, 161(1), 10. <https://doi.org/10.3847/1538-3881/abc3c8>
- Vousden, W. D., Farr, W. M., & Mandel, I. (2015). *Dynamic temperature selection for parallel tempering in Markov chain Monte Carlo simulations*. *Monthly Notices of the Royal Astronomical Society*, 455(2), 1919–1937. <https://doi.org/10.1093/mnras/stv2422>
- Winterhalder, T. O., Mérand, A., Kammerer, J., Lacour, S., Nowak, M., et al. (2025). *Astrometric exomoon detection by means of optical interferometry*. *Astronomy and Astrophysics*. <https://doi.org/10.48550/ARXIV.2509.15304>

## All VLTI/GRAVITY Dates of Observation of $\beta$ Pic b

**Table 4:** All available VLTI/GRAVITY dates of observation of  $\beta$  Pic b. Asterisks (\*) denote the data set used in this study. Double asterisks (\*\*) denote data not publicly available yet (still in proprietary period).

| Date of Observation | Note |
|---------------------|------|
| 2016-09-10          |      |
| 2016-09-12          |      |
| 2016-09-13          |      |
| 2016-09-15          |      |
| 2016-09-17          |      |
| 2016-09-18          |      |
| 2016-09-19          |      |
| 2016-10-16          |      |
| 2016-10-18          |      |
| 2016-10-20          |      |
| 2017-10-03          |      |
| 2017-12-04          |      |
| 2018-01-12          |      |
| 2018-09-01          |      |
| 2018-09-22          |      |
| 2018-11-25          |      |
| 2018-11-26          |      |
| 2019-09-12          | *    |
| 2019-11-09          | *    |
| 2019-11-11          | *    |
| 2020-01-07          | *    |
| 2020-02-09          | *    |
| 2020-02-10          | *    |
| 2020-02-12          | *    |
| 2020-03-08          | *    |
| 2020-03-12          | *    |
| 2021-01-06          |      |
| 2021-01-07          |      |
| 2021-08-27          |      |
| 2021-08-28          |      |
| 2022-01-25          |      |
| 2022-08-18          |      |
| 2022-10-13          |      |
| 2022-10-14          |      |
| 2022-10-15          |      |
| 2024-12-20          | **   |
| 2025-12-11          | **   |